

Homework 2 – Marine Chemistry

Total 10 points

Due date: Tuesday 07/12/2022 at 2 pm (class time)

- 1- What is meant by the statement “Water is familiar, abundant, and odd”?
 - a. What is the water molecules’ characteristic responsible for its “odd” and especial properties?
 - b. What are the “odd” properties of water? (2 Pts).

- 2- Using the north-south temperature distribution below, identify the thermocline (or pycnocline) and the deep zones at 50S, 0, and 40N. Negative values indicate the South Atlantic.

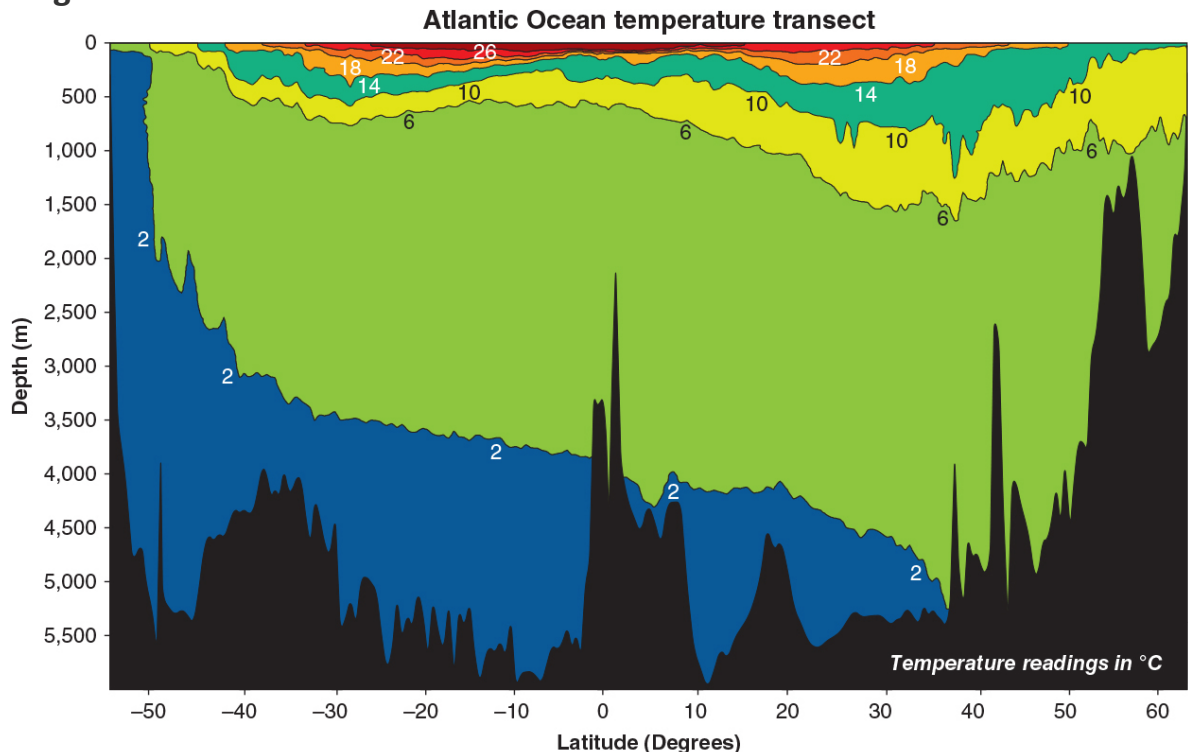


Figure 6.20 Temperature and salinity across the Atlantic Ocean basin at about 30° Degrees west longitude. Most seawater lies in cold darkness. In this image, a small and thin warm layer, concentrated near the equator, stands out above the relatively cool and consistently stable deep waters. As Earth warms, this layer will extend poleward. The irregular black shapes at the bottom of the graphs represent the ocean floor at the latitude shown.

- a. Is the thermocline more prominent in lower or higher latitudes?
- b. Based on the ice cube experiment video and your answers so far, in which region surface waters can reach the deep zones, higher or lower latitudes? (2 Pts).

- 3- Why is the water's high heat capacity important for local weather and global climate (2 pts)**
- 4- What are the major gases dissolved in the oceans? Are the amount of dissolved gases the same at the surface and the deep ocean? Which gas is more abundant in both regions? (2 pts)**
- 5- Based on the discussion of Giltz & Taylorf (2017) (2 pts):**
- a. Describe how a change in ocean pH can affect some marine organisms? Ex: corals**
 - b. Why are larval crustaceans (such as blue crab zoea) more susceptible to ocean acidification than juveniles and adults?**
 - c. What scientific evidence convinced you that these organisms were affected by ocean acidification?**
 - d. Should anthropogenic CO₂ be considered a marine pollutant?**
 - e. If yes, of what kind (chronic or acute) and why.**